Do Now · retrieval

Lesson 127 · Lossless and Lossy Compression

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is indirect addressing in a CPU instruction?

Q2

What is the binary result of 1 + 1 in a full adder with carry-in 0?

Q3

What is O(n) search complexity associated with?

Q4

Which memory is typically volatile: RAM or ROM?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find redundant data — Repeated patterns and predictable structures can be encoded efficiently.

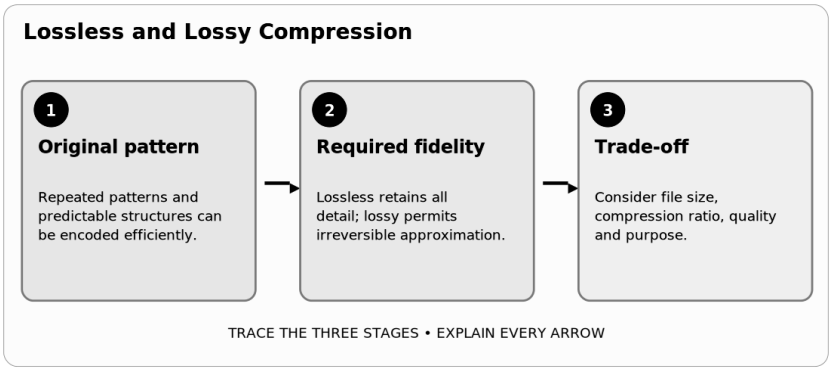
Core idea 2

Choose method — Lossless retains all detail; lossy permits irreversible approximation.

Core idea 3

Measure impact — Consider file size, compression ratio, quality and purpose.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

lossless — Original data can be reconstructed exactly.

lossy — Some original information is deliberately discarded.

redundancy — Repeated or predictable data.

compression ratio — Original size divided by compressed size (by this convention).

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A medical image must remain identical after decompression. Choose a technique and justify.

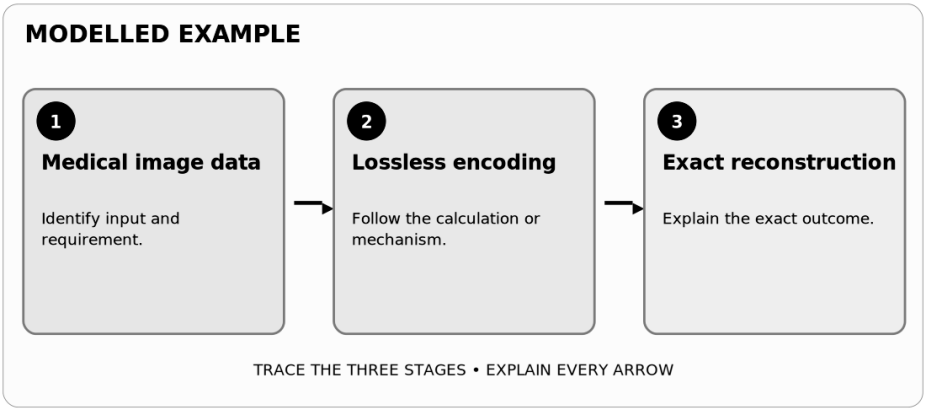
Follow the worked method

1. Medical image data — Predict the first step from the prompt.

2. Lossless encoding — Explain the intermediate result or decision.

3. Exact reconstruction — Lossless compression is required if exact original image data must be preserved. Lossy compression permanently discards information and could alter diagnostically important features.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Limited bandwidth
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define lossy and lossless compression and give one application of each. [6]

Question 2

A file is reduced from 12 MB to 3 MB. Calculate the original-to-compressed compression ratio. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate a compression choice for source code, an online photograph and an MRI scan. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A lossy method always reconstructs the original bit-for-bit.

Use key vocabulary from memory

1. Explain lossless in your own words.

2. Explain lossy in your own words.

3. Explain redundancy in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.